

1-Conformal predictive interval for right-censored survival data under finite observation window

Abstract

[placeholder]

1 Setup

We consider the construction of conformal predictive intervals for right-censored survival data under a finite observation window. Let T denote the true event time and $\tau < \infty$ a fixed study horizon. We define the truncated event time

$$T^* = \min(T, \tau),$$

which takes values in $[0, \tau]$ and represents the event time observable within the follow-up window. Our inferential target is a $(1 - \alpha)$ predictive interval for T^* given covariates: for a new subject with covariate vector z_0 , estimate

$$\hat{C}_{1-\alpha}(z_0) = [\hat{L}_\alpha(z_0), \hat{U}_\alpha(z_0)]$$

such that the conformal marginal coverage criterion

$$\Pr\{T^* \in \hat{C}_{1-\alpha}(Z)\} \geq 1 - \alpha$$

is achieved approximately.

To capture both non-administrative and administrative censoring behavior, let $\tilde{C}_i$ be a latent random censoring time and define the observed censoring variable under finite horizon by

$$C_i = \min(\tilde{C}_i, \tau).$$

Hence C_i is a mixed random variable with a point mass at τ . Let $(Y_i, \Delta_i, Z_i)_{i=1}^n$ denote independent observations, where

$$Y_i = \min(T_i, C_i) = \min(T_i^*, C_i), \quad \Delta_i = I(T_i \leq C_i),$$

and $Z_i \in \mathbb{R}^p$ is a vector of covariates. We assume independent censoring, $\tilde{C} \perp (T, Z)$.

2 Method

Direct application of conformal inference is impeded by censoring. Following the general strategy of resampling-based conformal calibration in [Qin et al. \(2025\)](#), we approximate the joint distribution of (T^*, Z) by a nonparametric estimator and generate synthetic samples for calibration.

[Qin et al. \(2025\)](#) define

$$F_T(t, z) = \Pr(T \leq t, Z \leq z), \quad G(t) = \Pr(C \geq t).$$

In the original right-censored setting (targeting (T, Z)), the IPCW (inverse probability of censoring weighted) estimator takes the form

$$\hat{F}_T(t, z) = \frac{1}{n} \sum_{i=1}^n \frac{\Delta_i I(Y_i \leq t, Z_i \leq z)}{\hat{G}(Y_i^-)}.$$

Now our target is the transformed variable $T^* = \min(T, \tau)$:

$$F(t, z) = \Pr(T^* \leq t, Z \leq z), \quad t \in [0, \tau].$$

To derive a nonparametric estimator of this $F(t, z)$ on (T^*, Z) , we consider two cases.

Case 1: $t < \tau$.

$$\begin{aligned} F(t, z) &= \Pr(\min(T, \tau) \leq t, Z \leq z) \\ &= \Pr(T \leq t, Z \leq z), \end{aligned}$$

because $\min(T, \tau) \leq t < \tau$ implies $T \leq t$. So the estimator is the same IPCW form as for F_T .

Case 2: $t = \tau$.

$$\begin{aligned} F(\tau, z) &= \Pr(\min(T, \tau) \leq \tau, Z \leq z) \\ &= \Pr(Z \leq z) \\ &= \Pr(T \leq \tau, Z \leq z) + \Pr(T > \tau, Z \leq z). \end{aligned}$$

Therefore, relative to the original (T, Z) target, $F(\tau, z)$ has an extra mass-at- τ component $\Pr(T > \tau, Z \leq z)$. Under $C = \min(C, \tau)$, records with $(Y = \tau, \Delta = 0)$ correspond to this component, and it is estimated by reweighting with $G(\tau^-)$:

$$\Pr(T > \tau, Z \leq z) = E \left[\frac{I(Y = \tau, \Delta = 0, Z \leq z)}{G(\tau^-)} \right].$$

Hence, for $t < \tau$:

$$\hat{F}(t, z) = \frac{1}{n} \sum_{i=1}^n \frac{\Delta_i I(Y_i \leq t, Z_i \leq z)}{\hat{G}(Y_i^-)}.$$

And at $t = \tau$:

$$\hat{F}(\tau, z) = \frac{1}{n} \sum_{i=1}^n \frac{\Delta_i I(Y_i \leq \tau, Z_i \leq z)}{\hat{G}(Y_i^-)} + \frac{1}{n} \sum_{i=1}^n \frac{I(Y_i = \tau, \Delta_i = 0, Z_i \leq z)}{\hat{G}(\tau^-)}.$$

Equivalently, on $t \in [0, \tau]$,

$$\hat{F}(t, z) = \frac{1}{n} \sum_{i=1}^n \frac{\Delta_i I(Y_i \leq t, Z_i \leq z)}{\hat{G}(Y_i^-)} + \frac{1}{n} \sum_{i=1}^n \frac{I(Y_i = \tau, \Delta_i = 0) I(t = \tau, Z_i \leq z)}{\hat{G}(\tau^-)}.$$

where the first term estimates the continuous part from observed failures, and the second term recovers the point mass at $T^* = \tau$. Because C has mass at τ , we emphasize that $G(\tau^-) = \Pr(C \geq \tau)$ is used rather than $\Pr(C > \tau)$, to avoid a zero denominator in KM estimation implementation.

To construct the predictive interval, we adopt the Cox proportional hazards model as a working model for T^* :

$$\lambda(t | z) = \lambda_0(t) \exp(z^\top \beta), \quad S(t | z) = \exp\{-\Lambda_0(t) \exp(z^\top \beta)\},$$

where $\Lambda_0(t) = \int_0^t \lambda_0(u) du$. Define the monotone pivot

$$U = S(T^* | Z),$$

which is strictly decreasing in T^* and therefore invertible.

2.1 Algorithm

Algorithm.

1. Fit the Cox proportional hazards model on the observed data to obtain estimators $\hat{\beta}$ and $\hat{\Lambda}_0(t)$.
2. Draw B synthetic samples $(T^{*(b)}, Z^{(b)})$ i.i.d. from the estimated joint distribution $\hat{F}$.
3. For each draw, compute the pivot

$$U^{(b)} = \exp\{-\hat{\Lambda}_0(T^{*(b)}) \exp(Z^{(b)\top} \hat{\beta})\}.$$

4. Let L and R be the empirical $\alpha/2$ and $(1 - \alpha/2)$ quantiles of $\{U^{(b)}\}_{b=1}^B$.
5. For a new covariate value z_0 , solve

$$L \leq \exp\{-\hat{\Lambda}_0(t) \exp(z_0^\top \hat{\beta})\} \leq R$$

to obtain the predictive interval

$$\hat{L}_\alpha(z_0) = \inf\{t : \exp[-\hat{\Lambda}_0(t) \exp(z_0^\top \hat{\beta})] \leq R\}, \quad \hat{U}_\alpha(z_0) = \sup\{t : \exp[-\hat{\Lambda}_0(t) \exp(z_0^\top \hat{\beta})] \geq L\}.$$

The final interval is truncated to $[0, \tau]$.

Algorithm 1 Conformal Predictive Interval for T^* under Finite Observation Window

Require: Observed data $D = \{(Y_i, \Delta_i, Z_i)\}_{i=1}^n$, miscoverage level α , calibration draws B , horizon τ .

Target: $(1 - \alpha)$ predictive interval for $T^* = \min(T, \tau)$.

- 1: **procedure** CONFORMALPI(D, α, B)
- 2: Fit Cox model to obtain $\hat{\beta}$ and $\hat{\Lambda}_0(t)$.
- 3: **for** $b = 1, \dots, B$ **do**
- 4: Draw $(T^{*(b)}, Z^{(b)}) \sim \hat{F}$.
- 5: Compute

$$U^{(b)} = \exp\{-\hat{\Lambda}_0(T^{*(b)}) \exp(Z^{(b)\top} \hat{\beta})\}.$$
- 6: **end for**
- 7: Let (L, R) be the empirical $(\alpha/2, 1 - \alpha/2)$ quantiles of $\{U^{(b)}\}_{b=1}^B$.
- 8: **Output:** $(\hat{\Lambda}_0, \hat{\beta}, L, R)$.
- 9: **end procedure**

10: **function** PREDICTTIME(z_0)

11: Solve

$$L \leq \exp\{-\hat{\Lambda}_0(t) \exp(z_0^\top \hat{\beta})\} \leq R$$

for $t \in [0, \tau]$.

12: **Return:** predictive interval $[\hat{L}_\alpha(z_0), \hat{U}_\alpha(z_0)]$.

13: **end function**

Remark (Lower bound for T). After obtaining the predictive interval $[\hat{L}_\alpha(z_0), \hat{U}_\alpha(z_0)]$ for $T^* = \min(T, \tau)$, we also get a one-sided lower bound for the original event time T . Because $T \geq T^*$ almost surely, the event $\{T^* \geq \hat{L}_\alpha(Z)\}$ implies $\{T \geq \hat{L}_\alpha(Z)\}$. Also,

$$\{\hat{L}_\alpha(Z) \leq T^* \leq \hat{U}_\alpha(Z)\} \subseteq \{T^* \geq \hat{L}_\alpha(Z)\}.$$

Hence

$$\Pr\{T \geq \hat{L}_\alpha(Z)\} \geq \Pr\{\hat{L}_\alpha(Z) \leq T^* \leq \hat{U}_\alpha(Z)\} \geq 1 - \alpha.$$

So $\hat{L}_\alpha(z_0)$ can be interpreted as a conservative lower prediction bound for T .

Extension (Fixed-horizon risk). Besides outputting an interval for T^* , for a new unit with covariate z_0 we want a single number at a fixed horizon $t_0 \in [0, \tau]$ that summarizes “risk by t_0 ”, which means the probability of event occurring before t_0 . To estimate this probability,

Define

$$u_0 = \hat{S}(t_0 \mid z_0) = \exp\{-\hat{\Lambda}_0(t_0) \exp(z_0^\top \hat{\beta})\}.$$

Using the same Monte Carlo draws from the main algorithm,

$$U^{(b)} = \exp\{-\hat{\Lambda}_0(T^{*(b)}) \exp(Z^{(b)\top} \hat{\beta})\}, \quad (T^{*(b)}, Z^{(b)}) \sim \hat{F},$$

we define

$$\hat{\pi}_{\text{marg}}(t_0, z_0) = \frac{1}{B} \sum_{b=1}^B I\{U^{(b)} \geq u_0\}.$$

This estimates

$$\pi_{\text{marg}}(t_0, z_0) := \Pr\{U \geq u_0\}, \quad U = \hat{S}(T^* \mid Z), \quad (T^*, Z) \sim \hat{F}.$$

So, consistent with the interval construction, z_0 enters through the threshold u_0 , while calibration is with respect to the estimated joint distribution $\hat{F}$. This is a marginally calibrated, z_0 -adaptive probability, not a strictly conditional probability. For fixed z_0 , since u_0 decreases as t_0 increases, $\hat{\pi}_{\text{marg}}(t_0, z_0)$ is nondecreasing in t_0 .

3 Simulation study

3.1 Simulation design and sensitivity settings

We evaluate the finite-window conformal predictive interval under a factorial simulation design with one baseline scenario and three sensitivity axes: (i) stronger covariate effects, (ii) higher Weibull shape, and (iii) higher covariate dimension. The event time is generated from a Weibull proportional hazards model:

$$h(t \mid Z) = \frac{\gamma}{\kappa} \left(\frac{t}{\kappa}\right)^{\gamma-1} \exp(Z^\top \beta), \quad T = \kappa \left\{ \frac{-\log U}{\exp(Z^\top \beta)} \right\}^{1/\gamma}, \quad U \sim \text{Unif}(0, 1).$$

For each replicate, we generate independent covariates $Z = (Z_1, \dots, Z_p)$ with $Z_j \sim N(0, 1)$, define τ as a fixed quantile of generated T , generate latent censoring $\tilde{C} \sim \text{Exp}(\lambda_c)$, then set $C = \min(\tilde{C}, \tau)$. The observed data are (Y, Δ) with $Y = \min(T, C)$ and $\Delta = I(T \leq C)$, and the prediction target is $T^* = \min(T, \tau)$.

Table 1: Simulation settings and sensitivity factors.

Component	Setting
Failure-time DGP	Weibull PH with $\kappa = 1$; shape $\gamma \in \{1.2, 1.8\}$ (baseline $\gamma = 1.2$).
Covariates	Independent normal covariates; baseline dimension $p = 3$, high-dimension sensitivity $p = 9$.
Regression effects	$\beta_j = 0.30$ (baseline); stronger effect $\beta_j = 0.60$.
Finite horizon τ	τ set to quantiles of generated T : 0.75, 0.90, 0.97.
Censoring	$\tilde{C} \sim \text{Exp}(\lambda_c)$, $C = \min(\tilde{C}, \tau)$, with target non-administrative censoring 10%, 20%, 35%.
Sample size and split	$n = 2000$ per replicate; 70% training and 30% testing.
Conformal calibration	Cox working model; Monte Carlo draws $B = 1500$; miscoverage $\alpha = 0.1$.
Replicates	30 replicates per setting (1080 total).

The rate λ_c is calibrated to target approximately 10%, 20%, or 35% non-administrative censoring before τ . For each setting, we simulate $n = 2000$ observations and split the data into training/testing subsets with proportion 70%/30%. The conformal method uses a Cox working model and Monte Carlo calibration with $B = 1500$ draws at $\alpha = 0.1$. Each setting is repeated 30 times, giving $4 \times 3 \times 3 = 36$ settings and 1080 total simulation replicates.

We report empirical coverage and mean interval length on the test set:

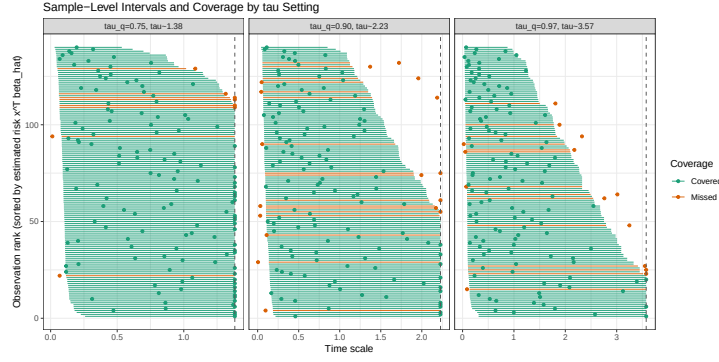
$$\widehat{\text{Cov}} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} I\{\widehat{L}_i \leq T_i^* \leq \widehat{U}_i\}, \quad \widehat{\text{Len}} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} (\widehat{U}_i - \widehat{L}_i).$$

3.2 Simulation results

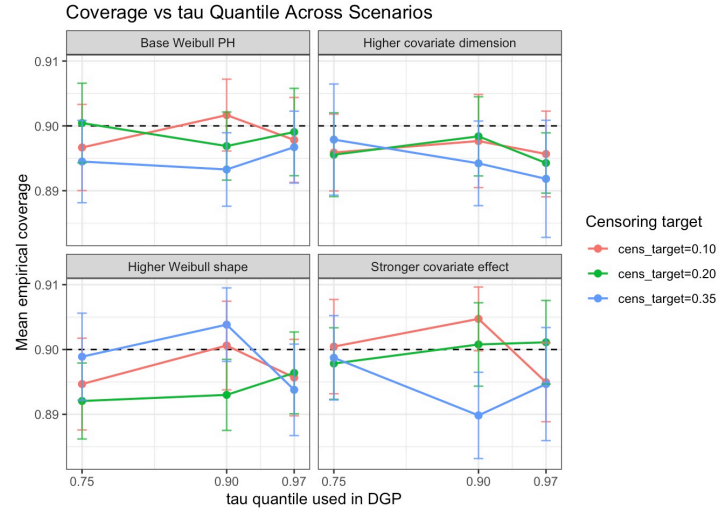
Across all settings, the overall mean empirical coverage is 0.897 for target 0.900, indicating good calibration under finite-window truncation.

Figure 1 presents the simulation results. In panel (a), each horizontal line segment is one predicted interval $[\widehat{L}_i, \widehat{U}_i]$ for a randomly sampled subset (140 subjects) from the test set, ordered by estimated risk score $x^\top \widehat{\beta}$, and the dot on that segment is the corresponding true value T_i^* . Green means covered and orange means missed. The vertical dashed line marks the realized τ in that facet, and each facet corresponds to one τ -quantile regime. In panel (b), the x -axis is the τ quantile used in data generation and the y -axis is mean empirical coverage; colors indicate censoring targets, error bars are 95% CIs across replicates, and the dashed horizontal line is nominal coverage 0.9. In panel (c), the structure is the same as panel (b), but the y -axis is mean interval length instead of coverage. In panel (d), each point is one simulation setting; the x -axis is mean length, the y -axis is mean coverage, point color encodes censoring target, and point shape encodes τ setting.

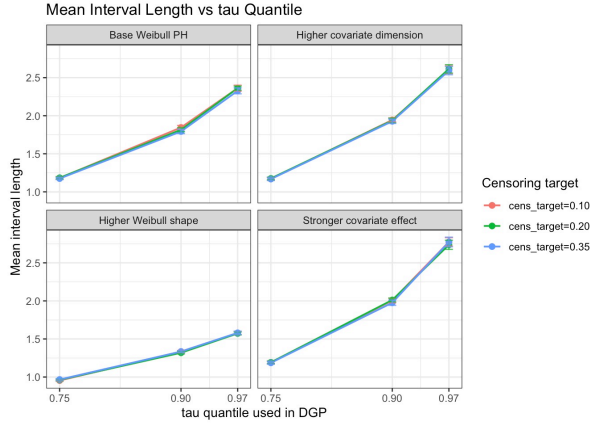
From the plots, panel (a) shows that most intervals cover their corresponding T^* , and interval positions and widths shift as τ changes. Panel (b) shows coverage remains close to nominal across the three τ regimes and censoring targets, with only mild under-coverage in some high-censoring and large- τ settings. Panel (c) shows interval length increases as τ moves from the 0.75 quantile to the 0.97 quantile, which is expected because the range of T^* becomes wider. Panel (d) summarizes the coverage-length tradeoff: higher Weibull shape tends to shorten intervals, whereas strong signal and higher dimension lead to longer intervals.



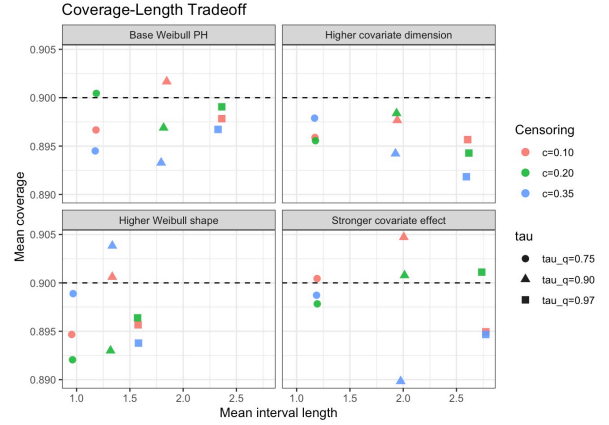
(a) Empirical interval coverage examples.



(b) Coverage versus τ quantile.



(c) Mean interval length versus τ quantile.



(d) Coverage-length tradeoff.

Figure 1: Simulation results for finite-window conformal predictive intervals under varying DGP, censoring targets, and τ settings. (a) Direct sample-level intervals and coverage status. (b) Coverage versus τ quantile with nominal line and replicate-level uncertainty bars. (c) Mean interval length versus τ quantile. (d) Coverage-length tradeoff across settings.

Averaging over censoring and τ settings, mean coverage is very similar across DGP variants (baseline: 0.897, strong signal: 0.898, high shape: 0.897, high dimension: 0.896). This indicates that empirical coverage is close to nominal and remains stable across different DGPs, while interval length differs more noticeably across parameter settings.

3.3 Baseline comparison with a Cox plug-in interval

To examine whether dropping conformal calibration yields narrower intervals, we add a non-conformal benchmark that targets the same truncated event time $T^* = \min(T, \tau)$. The benchmark fits the same Cox working model on the training data, uses the fitted survival curve $\hat{S}(t | z)$ to obtain the implied predictive distribution of T^* , and then outputs the equal-tailed interval

$$[\hat{Q}_{T^*|z}(\alpha/2), \hat{Q}_{T^*|z}(1 - \alpha/2)]$$

without conformal calibration.

For a fair comparison, we keep the baseline setting fixed: $n = 2000$, $p = 3$, $\beta_j = 0.30$, $\gamma = 1.2$, $\kappa = 1$, τ at the 0.90 quantile of T , target non-administrative censoring 0.20, $\alpha = 0.10$, and the same 70%/30% train-test split. The conformal method uses $B = 1500$ Monte Carlo draws, while the Cox plug-in method uses only the fitted Cox model. The comparison is repeated 20 times.

Figure 2 shows that the Cox plug-in interval is not narrower in this baseline setting. In panel (a), the plug-in intervals are often slightly longer, especially near the upper end of the time axis. Panel (b) shows the same pattern after aggregation: mean empirical coverage is about 0.893 for the conformal method and 0.928 for the Cox plug-in benchmark, while mean interval length is about 1.814 and 1.975, respectively. Panel (c) confirms this at the sample level: most points lie above the diagonal, indicating longer Cox plug-in intervals for the same test units. So, at least in this baseline experiment, removing conformal calibration does not lead to tighter predictive intervals.

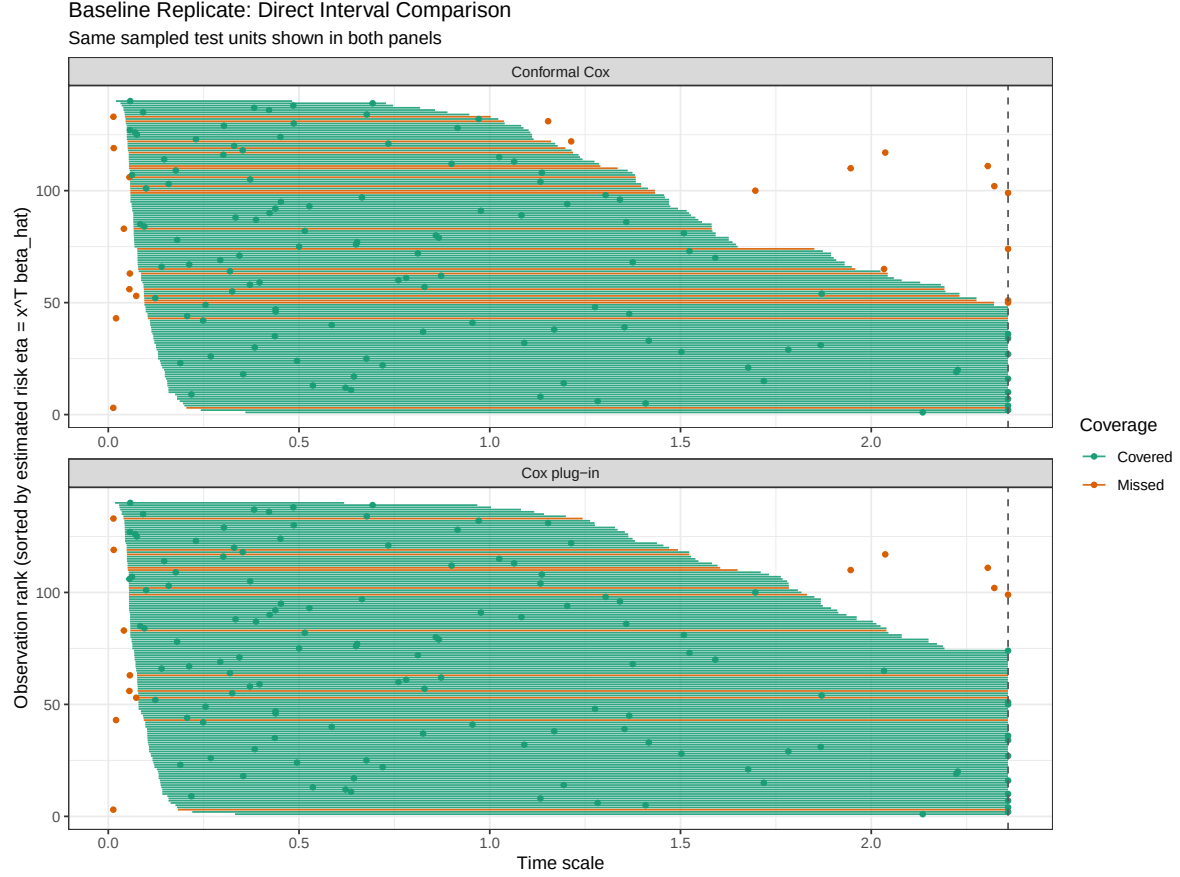
3.4 Simulation for the fixed-horizon risk extension

We additionally assess the extension in the baseline setting ($n = 2000$, $p = 3$, $\beta_j = 0.30$, $\gamma = 1.2$, $\kappa = 1$, τ at the 0.90 quantile of T , censoring target 0.20, $\alpha = 0.10$, $B = 1500$). Here $\eta := x^\top \hat{\beta}$ denotes the estimated linear predictor from the working Cox model. For one baseline replicate, we select four test units with different estimated risk scores $x^\top \hat{\beta}$ (approximately low to high), and compute $\hat{\pi}_{\text{marg}}(t_0, z)$ on a grid of $t_0 \in [0, \tau]$.

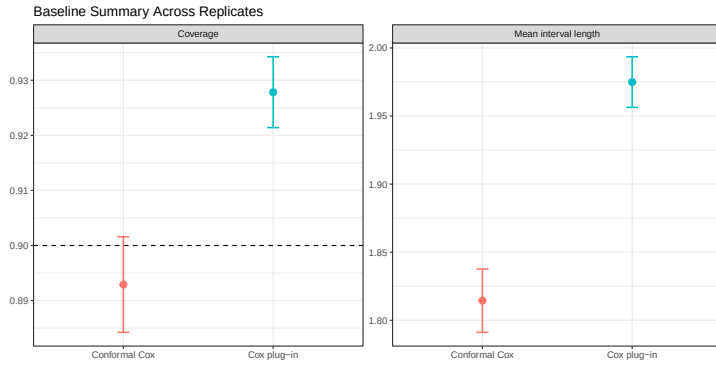
Figure 3(a) shows that, for each selected unit, $\hat{\pi}_{\text{marg}}(t_0, z)$ increases with t_0 , as expected. Across units, the curves are ordered by risk level: units with larger $x^\top \hat{\beta}$ tend to have higher risk-score values at the same t_0 . Figure 3(b) provides a reference of the four samples with regard to $\eta = x^\top \hat{\beta}$ distribution among the test set.

References

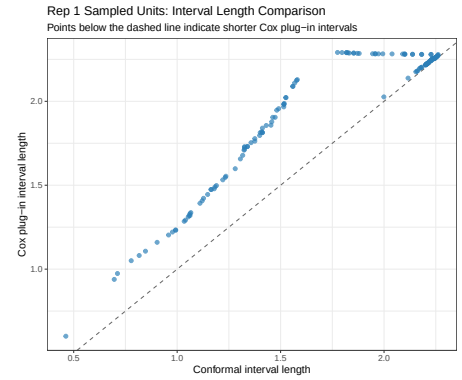
Qin, J., Piao, J., Ning, J., and Shen, Y. (2025). Conformal predictive intervals in survival analysis: a resampling approach. *Biometrics*, 81(2):ujaf063.



(a) Direct comparison on the same 140 sampled test units from one baseline replicate, ordered by $\eta = x^T \hat{\beta}$.

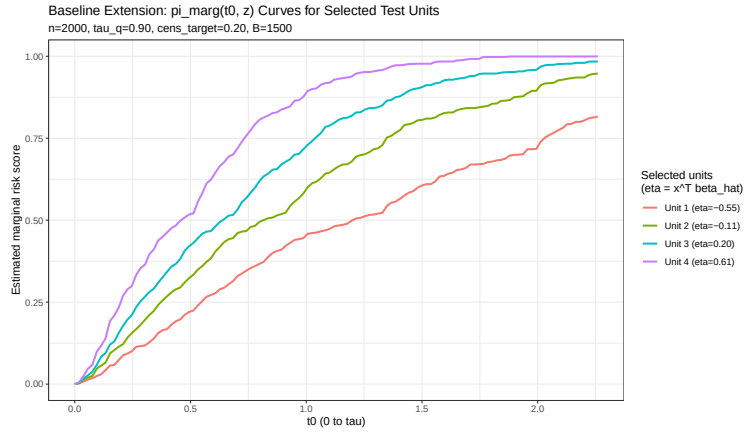


(b) Aggregate coverage and mean interval length across 20 replicates.

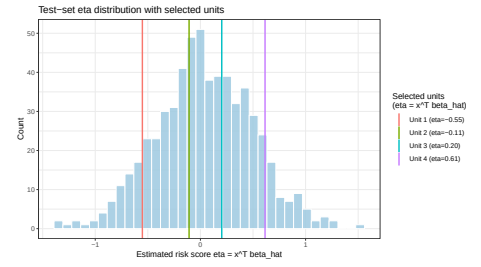


(c) Replicate-1 paired interval lengths for the same sampled units.

Figure 2: Baseline comparison between conformal Cox intervals and Cox plug-in intervals without conformal calibration. In panel (a), points are realized T^* values and horizontal segments are predicted intervals under the two methods for the same sampled test subjects. In panel (b), points and error bars summarize replicate means and 95% confidence intervals. In panel (c), points below the diagonal correspond to shorter Cox plug-in intervals.



(a) $\hat{\pi}_{\text{marg}}(t_0, z)$ curves for four selected test units.



(b) Test-set $\eta = x^T \hat{\beta}$ distribution and selected units.

Figure 3: Baseline simulation for the fixed-horizon risk-score extension.